BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . MEDMASTERS COLLABORATIVE

Module 11: Blood

Week 14 teaching guide . 2 topics

This is the instructor-facing companion to the student pre-work hub. For every topic, you have the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Lecture order, common misconceptions, and pre-video drawing prompts are baked in so each video lesson can hit the same pedagogical beats.

Topics in this module

Module 11 of 17 . Dr. Sharilyn Rennie

TOPIC 1 OF 2 . WEEK 14 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Blood Composition and Hemopoiesis

Plasma, formed elements, and how blood cells are made in red marrow.

The Science

Blood is connective tissue: cells in a liquid matrix (plasma). About 55% plasma, 45% formed elements by volume. Plasma is mostly water plus proteins (albumin maintains oncotic pressure; globulins include antibodies; fibrinogen makes fibrin for clots), electrolytes, glucose, hormones, and wastes.

Formed elements: erythrocytes (RBCs, 99% of cells; biconcave, no nucleus, packed with hemoglobin, 120-day lifespan, function = O_2/CO_2 transport). Leukocytes (WBCs, immune cells). Platelets (cell fragments from megakaryocytes; central to hemostasis).

WBC types: granulocytes (neutrophils, eosinophils, basophils, granular cytoplasm) and agranulocytes (lymphocytes, monocytes). Neutrophils are the most abundant; first responders to bacterial infection. Eosinophils target parasites and allergic responses. Basophils release histamine in inflammation. Lymphocytes (B and T) drive adaptive immunity. Monocytes leave blood and mature into tissue macrophages.

Hematopoiesis: all blood cells come from hematopoietic stem cells in red marrow (axial skeleton and proximal limb bones in adults). Erythropoiesis (RBC production) is driven by erythropoietin, made by the kidneys in response to low oxygen. Low EPO (chronic kidney disease) = anemia.

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the percentages of plasma vs cells, and to list the five kinds of WBC in order of abundance. They have to commit first.

Common misconceptions to address

- Students think blood is just RBCs. By cell count yes, but by function the WBCs and platelets are equally critical.
- Blood is sometimes described as 'red liquid.' It is a connective tissue with a specific cell-to-matrix structure.
- RBCs without a nucleus seems like a disability. It is a specialization: more room for hemoglobin, more flexible to squeeze through capillaries.

Order of operations (what to teach first)

Plasma vs formed elements, then RBC specifics, then WBC types and roles, then platelets briefly, then hematopoiesis.

Self-test prompts (for students before the recall deck)

- What percentage of blood is plasma?
- Name the WBC types from most to least abundant.

- Why are patients with chronic kidney disease anemic?

Why This Matters to Your Patients

CBC (complete blood count) is the most common lab test. Low hemoglobin = anemia (many causes: iron deficiency, B12, chronic disease, blood loss). High WBC = infection or leukemia. Low platelets = bleeding risk. The differential (percentages of each WBC type) tells you what kind of infection: bacterial typically raises neutrophils; viral often raises lymphocytes; parasites raise eosinophils. Patients on chemotherapy have predictable cytopenias from bone marrow suppression; nurses time their assessments and protective measures accordingly. Recombinant EPO is given to dialysis patients to treat anemia of CKD.

TOPIC 2 OF 2 . WEEK 14 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hemostasis and Blood Typing

Stopping bleeding in three steps, and the basics of ABO and Rh.

The Science

Three phases of hemostasis: vascular spasm (damaged vessel constricts), platelet plug formation (platelets adhere to exposed collagen via von Willebrand factor, activate, and recruit more platelets), and coagulation (the clotting cascade ends in fibrin, which reinforces the plug). The cascade has intrinsic and extrinsic pathways that converge on a common pathway producing thrombin, which converts fibrinogen to fibrin.

ABO typing: RBCs may carry A antigen, B antigen, both, or neither (Type O). Plasma carries antibodies against absent antigens (Type A has anti-B; Type O has both; Type AB has neither). Type O is the universal RBC donor (no antigens to trigger recipient antibodies); Type AB is the universal RBC recipient (no antibodies to attack donor RBCs). For plasma, the rule reverses: AB is the universal plasma donor.

Rh factor: Rh+ means D antigen present. Anti-Rh antibodies form only after exposure. Critical in pregnancy: an Rh- mother carrying an Rh+ fetus may produce anti-Rh antibodies, which can cross the placenta in a future pregnancy and attack an Rh+ fetus, producing hemolytic disease of the newborn. RhoGAM (anti-Rh antibody given to the mother) prevents this by clearing fetal Rh+ cells before her immune system responds.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the three phases of hemostasis as a flowchart and predict what a missing platelet plug, or missing fibrin, would mean.

Common misconceptions to address

- Students think clotting and clot dissolution are separate things. They are balanced: fibrinolysis is happening alongside coagulation; the net result is the lifespan of the clot.
- ABO and Rh are sometimes mixed. ABO antibodies are pre-formed (natural); Rh antibodies form only after exposure.
- Type O is universal donor for RBCs, NOT for plasma.

Order of operations (what to teach first)

Three phases of hemostasis, then ABO logic, then Rh logic with the pregnancy implication.

Self-test prompts (for students before the recall deck)

- Name the three phases of hemostasis.
- Which is the universal RBC donor and which is the universal plasma donor?
- Why are Rh- women given RhoGAM during pregnancy?

Why This Matters to Your Patients

Hemophilia A and B (clotting factor deficiencies) produce delayed bleeding into joints and tissue. Von Willebrand disease (the most common inherited bleeding disorder) produces mucocutaneous bleeding. Warfarin inhibits vitamin-K-dependent clotting factors and is used to prevent stroke in atrial fibrillation; the trade-off is bleeding risk, which is monitored with INR. Heparin inhibits thrombin and is used for acute anticoagulation. Transfusion reactions occur when ABO mismatch happens (recipient antibodies attack donor RBCs); hemolysis can be rapidly fatal, hence the two-nurse identity check at the bedside. Liver disease produces coagulopathy because the liver makes most clotting factors.